

MLB's called strike zone: a failed placebo and two contractions

Introduction

Comparing framing or pitch location across seasons means accounting for changes in what umpires call strikes. In 2025 the umpire grading buffer narrowed from two inches outside the edge to three-quarters of an inch either side (Andrews, FanGraphs, 5 May 2025); in 2026 regular-season ABS challenges began. I measured both steps against three old-rule seasons on one measurement plane, with intervals; I found no published estimate that does.

Methods

I re-projected 2022-2025 pitches to Statcast's 2026 plate midpoint. The main height rule used roster height plus a fixed offset throughout. Sample: 1,830,231 called pitches in 12,060 regular-season games, 2022 through 21 September 2026. Near the edge, I fitted a binomial GAM to the umpire's original, pre-challenge call by location and season, with umpire effects partially pooled. Each season's zone is its 50 percent contour; model-conditional intervals (95% unless marked 90%) use 1,000 coefficient draws and omit season-to-season variation. With ABS-measured heights, the rebuilt geometry, not the fitted model, matched 10,164 of 10,168 ABS challenge verdicts. I measured both steps net of the 2022-2024 trend. The placebo pair, 2023 and 2024, had no change to the zone or umpire grading, although 2024 changed the clock and running lane.

Results

The pre-registered placebo failed: the zone grew 10.8 (90% CI 7.7 to 13.9) square inches, beyond its ± 3 margin, so every step is descriptive. Net of the trend, the area fell 23.4 (95% CI 18.6 to 28.3) square inches in 2025 and 33.1 (95% CI 28.7 to 37.5) in 2026 (Figure 1). The zone lost width mostly in 2025 and height mostly in 2026 (Table 1). Within 3 inches of the rule-book zone's edge, the called-strike rate fell 10.69 (95% CI 10.04 to 11.28) percentage points from 2024 to 2026. A published comparison (Clemens, FanGraphs, 28 April 2026) reads 2025 at the plate front and 2026 at mid-plate. That convention gives a loss of 20.0 (95% CI 13.0 to 28.2) square inches; with both at mid-plate it is 32.8 (95% CI 29.3 to 36.6). The binned estimator failed the ± 3 agreement tolerance, giving contractions of 29.5 and 40.7 square inches. Top-edge intervals covered 91 of 100 recovery runs, below the 93 bound.

Conclusion

The 2025 zone was already smaller than any 2022-2024 zone, before regular-season challenges began, so baselines that start from 2025 miss that step. On model-conditional intervals the 2026 step was the larger, 59 percent of the two combined (95% CI 53 to 65), but those intervals omit the season-to-season variation needed to rank the steps. Main limit: the failed placebo; the design cannot assign either step to its rule. The sealed-set definition and acceptance criteria were committed publicly before the sealed set was opened. Games from 22 September 2026 on remain sealed for a confirmatory calibration check after the postseason.

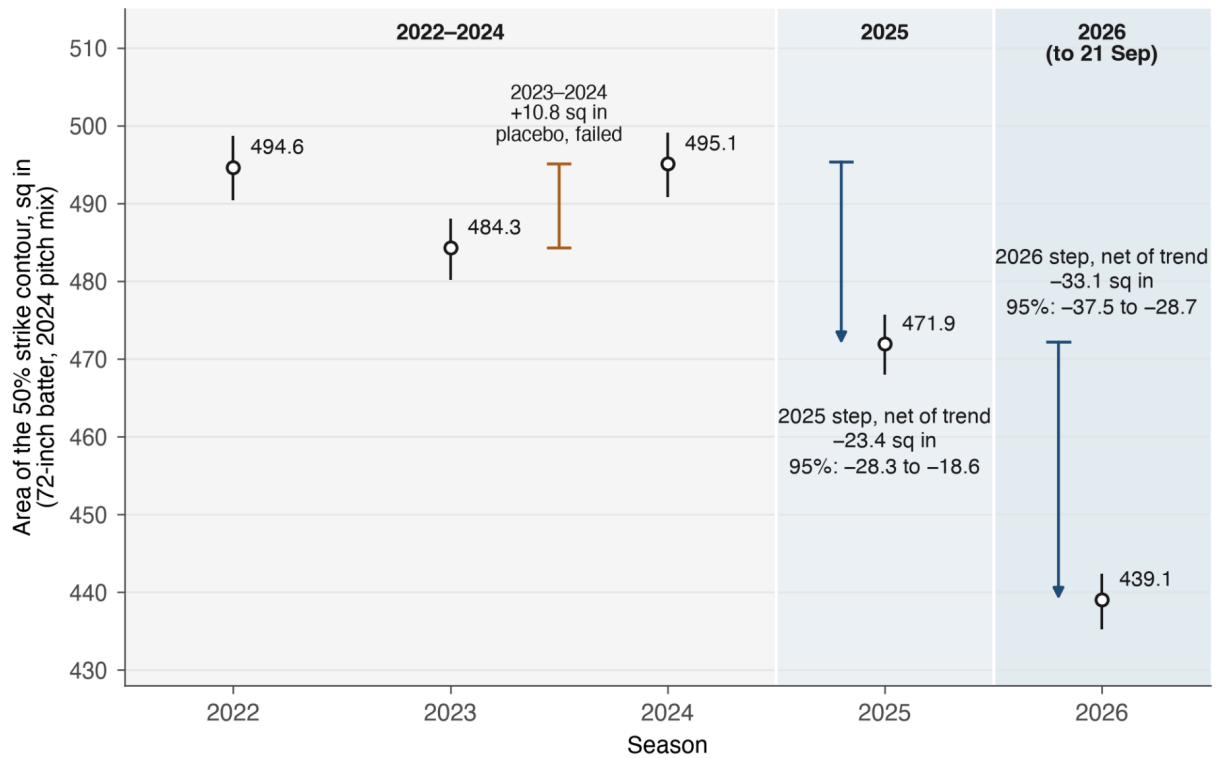


Figure 1. Points are fitted 50 percent contour areas (72-inch batter, 2024 pitch mix), with 95% model-conditional intervals. Arrows are the 2025 and 2026 steps net of the 2022–2024 trend of +0.3 sq in per season (95%: –1.6 to 2.2), each from the prior season plus that trend. The bracket is the 2023-to-2024 change, 10.8 sq in (90%: 7.7 to 13.9), with zone and grading rules unchanged; that placebo failed. Grading changed in 2025, regular-season challenges began in 2026; neither step is attributed. A sealed-set calibration check is pending.

Change in the fitted 50 percent contour	Top edge (in)	Bottom edge (in)	Half-width (in)	Area (sq in)	Shadow-band strike rate (pp)
2022-2024 trend, per season	0.13 [0.09, 0.18]	0.06 [0.02, 0.09]	-0.03 [-0.06, -0.01]	0.3 [-1.6, 2.2]	-0.09 [-0.41, 0.24]
2023 to 2024, zone rules unchanged (placebo, pre-registered on area and rate; margins ± 3 sq in and ± 0.5 pp): FAILED	0.47 [0.37, 0.57] 95% interval	0.16 [0.08, 0.24] 95% interval	0.12 [0.06, 0.18] 95% interval	10.8 [7.7, 13.9] 90% interval	1.54 [1.03, 2.04] 90% interval
2025 step, net of trend (grading buffer 0.75 in)	-0.18 [-0.32, -0.05]	0.18 [0.08, 0.29]	-0.41 [-0.49, -0.33]	-23.4 [-28.3, -18.6]	-4.46 [-5.28, -3.63]
2026 step, net of trend (ABS challenges)	-0.63 [-0.76, -0.50]	0.65 [0.55, 0.74]	-0.25 [-0.32, -0.16]	-33.1 [-37.5, -28.7]	-6.05 [-6.81, -5.31]
2024 to 2026, observed change (not net of trend)	-0.54 [-0.65, -0.43]	0.94 [0.85, 1.03]	-0.72 [-0.80, -0.64]	-56.0 [-59.6, -52.0]	-10.69 [-11.28, -10.04]
Measurement-plane adjustment, not a period change: 2025 read at mid-plate minus at the front	-0.77 [-0.94, -0.61]	-1.11 [-1.24, -0.98]	0.12 [-0.01, 0.25]	12.9 [6.3, 19.2]	—

Cells are point [95% interval] from 1,000 draws of the fitted model's coefficients, $N(\beta, V_c)$, conditional on the model and omitting season-to-season variation. The placebo row prints its pre-registered 90% interval on area and rate; its edge cells, added after the tag, are not pre-registered. Positive top or bottom values move that edge up; positive half-width values widen the zone; positive area values enlarge it. pp is percentage points; the shadow band is the pitches within 3 inches, either side, of where the ball just touches the zone edge. In the recovery test, top-edge 95% intervals covered the truth in 91 of 100 runs (bound 93). The placebo pair moved beyond its margin, so no step is attributed to its rule; the 2025 and 2026 labels name the period only. Dashes: not estimated for that row. Exploratory, not pre-registered: the 2022-to-2023 area change was -10.3 [-13.4, -7.3] sq in (90%); against the 2022-2024 mean plus one season of trend, the 2025 area step is -19.7 [-23.3, -16.1] sq in; against 2023 alone, -12.4 [-15.8, -8.5] sq in.

Table 1. Changes in the fitted 50 percent strike contour (72-inch batter, 2024 pitch mix, mid-plate), by period: the trend, the failed placebo pair, the two trend-adjusted steps, the unadjusted 2024-to-2026 change, and the measurement-plane adjustment. The note under the table defines the cells. A sealed-set calibration check is pending after the postseason.